

3.E Bockstein Homomorphisms

Homology and cohomology with coefficients in a field, particularly $\mathbb{Z}_p$ with p prime, often have more structure and are easier to compute than with $\mathbb{Z}$ coefficients. Of course, passing from $\mathbb{Z}$ to $\mathbb{Z}_p$ coefficients can involve a certain loss of information, a blurring of finer distinctions. For example, a $\mathbb{Z}_{p^n}$ in integer homology becomes a pair of $\mathbb{Z}_p$'s in $\mathbb{Z}_p$ homology or cohomology, so the exponent n is lost with $\mathbb{Z}_p$ coefficients. In this section we introduce Bockstein homomorphisms, which in many interesting cases allow one to recover $\mathbb{Z}$ coefficient information from $\mathbb{Z}_p$ coefficients. Bockstein homomorphisms also provide a small piece of extra internal structure to $\mathbb{Z}_p$ homology or cohomology itself, which can be quite useful.

We will concentrate on cohomology in order to have cup products available, but the basic constructions work equally well for homology. If we take a short exact sequence $0 \rightarrow G \rightarrow H \rightarrow K \rightarrow 0$ of abelian groups and apply the covariant functor $\text{Hom}(C_n(X), -)$, we obtain

$$0 \rightarrow C^n(X; G) \rightarrow C^n(X; H) \rightarrow C^n(X; K) \rightarrow 0$$

which is exact since $C_n(X)$ is free. Letting n vary, we have a short exact sequence of chain complexes, so there is an associated long exact sequence

$$\cdots \rightarrow H^n(X; G) \rightarrow H^n(X; H) \rightarrow H^n(X; K) \rightarrow H^{n+1}(X; G) \rightarrow \cdots$$

whose ‘boundary’ map $H^n(X; K) \rightarrow H^{n+1}(X; G)$ is called a **Bockstein homomorphism**.

We shall be interested primarily in the Bockstein $\beta: H^n(X; \mathbb{Z}_m) \rightarrow H^{n+1}(X; \mathbb{Z}_m)$ associated to the coefficient sequence $0 \rightarrow \mathbb{Z}_m \xrightarrow{m} \mathbb{Z}_{m^2} \rightarrow \mathbb{Z}_m \rightarrow 0$, especially when m is prime, but for the moment we do not need this assumption. Closely related to β is the Bockstein $\tilde{\beta}: H^n(X; \mathbb{Z}_m) \rightarrow H^{n+1}(X; \mathbb{Z})$ associated to $0 \rightarrow \mathbb{Z} \xrightarrow{m} \mathbb{Z} \rightarrow \mathbb{Z}_m \rightarrow 0$. From the natural map of the latter short exact sequence onto the former one, we obtain the relationship $\beta = \rho \tilde{\beta}$ where $\rho: H^*(X; \mathbb{Z}) \rightarrow H^*(X; \mathbb{Z}_m)$ is the homomorphism induced by the map $\mathbb{Z} \rightarrow \mathbb{Z}_m$ reducing coefficients mod m . Thus we have a commutative triangle in the following diagram, whose upper row is the exact sequence containing $\tilde{\beta}$.

$$\begin{array}{ccccccc} H^n(X; \mathbb{Z}) & \xrightarrow{\rho} & H^n(X; \mathbb{Z}_m) & \xrightarrow{\tilde{\beta}} & H^{n+1}(X; \mathbb{Z}) & \xrightarrow{m} & H^{n+1}(X; \mathbb{Z}) \\ & & & \searrow \beta & \downarrow \rho & & \\ & & & & H^{n+1}(X; \mathbb{Z}_m) & & \end{array}$$

Example 3E.1. Let X be a $K(\mathbb{Z}_m, 1)$, for example $\mathbb{R}P^\infty$ when $m = 2$ or an infinite-dimensional lens space with fundamental group $\mathbb{Z}_m$ for arbitrary m . From the homology calculations in Examples 2.42 and 2.43 together with the universal coefficient theorem or cellular cohomology we have $H^n(X; \mathbb{Z}_m) \approx \mathbb{Z}_m$ for all n . Let us show that $\beta: H^n(X; \mathbb{Z}_m) \rightarrow H^{n+1}(X; \mathbb{Z}_m)$ is an isomorphism for n odd and zero for n even. If n is odd the vertical map ρ in the diagram above is surjective for $X = K(\mathbb{Z}_m, 1)$, as

is $\tilde{\beta}$ since the map m is trivial, so β is surjective, hence an isomorphism. On the other hand, when n is even the first map ρ in the diagram is surjective, so $\tilde{\beta} = 0$ by exactness, hence $\beta = 0$.

A useful fact about β is that it satisfies the derivation property

$$(*) \quad \beta(a \smile b) = \beta(a) \smile b + (-1)^{|a|} a \smile \beta(b)$$

which comes from the corresponding formula for ordinary coboundary. Namely, let φ and ψ be $\mathbb{Z}_m$ cocycles representing a and b , and let $\tilde{\varphi}$ and $\tilde{\psi}$ be lifts of these to $\mathbb{Z}_{m^2}$ cochains. Concretely, one can view φ and ψ as functions on singular simplices with values in $\{0, 1, \dots, m-1\}$, and then $\tilde{\varphi}$ and $\tilde{\psi}$ can be taken to be the same functions, but with $\{0, 1, \dots, m-1\}$ regarded as a subset of $\mathbb{Z}_{m^2}$. Then $\delta\tilde{\varphi} = m\eta$ and $\delta\tilde{\psi} = m\mu$ for $\mathbb{Z}_p$ cocycles η and μ representing $\beta(a)$ and $\beta(b)$. Taking cup products, $\tilde{\varphi} \smile \tilde{\psi}$ is a $\mathbb{Z}_{m^2}$ cochain lifting the $\mathbb{Z}_m$ cocycle $\varphi \smile \psi$, and

$$\delta(\tilde{\varphi} \smile \tilde{\psi}) = \delta\tilde{\varphi} \smile \tilde{\psi} \pm \tilde{\varphi} \smile \delta\tilde{\psi} = m\eta \smile \tilde{\psi} \pm \tilde{\varphi} \smile m\mu = m(\eta \smile \psi \pm \varphi \smile \mu)$$

where the sign $\pm$ is $(-1)^{|a|}$. Hence $\eta \smile \psi + (-1)^{|a|} \varphi \smile \mu$ represents $\beta(a \smile b)$, giving the formula $(*)$.

Example 3E.2: Cup Products in Lens Spaces. The cup product structure for lens spaces was computed in Example 3.41 via Poincaré duality, but using Bocksteins we can deduce it from the cup product structure in $\mathbb{CP}^\infty$, which was computed in Theorem 3.12 without Poincaré duality. Consider first the infinite-dimensional lens space $L = S^\infty / \mathbb{Z}_m$ where $\mathbb{Z}_m$ acts on the unit sphere S^∞ in $\mathbb{C}^\infty$ by scalar multiplication, so the action is generated by the rotation $v \mapsto e^{2\pi i/m} v$. The quotient map $S^\infty \rightarrow \mathbb{CP}^\infty$ factors through L , so we have a projection $L \rightarrow \mathbb{CP}^\infty$. Looking at the cell structure on L described in Example 2.43, we see that each even-dimensional cell of L projects homeomorphically onto the corresponding cell of $\mathbb{CP}^\infty$. Namely, the $2n$ -cell of L is the homeomorphic image of the $2n$ -cell in $S^{2n+1} \subset \mathbb{C}^{n+1}$ formed by the points $\cos \theta(z_1, \dots, z_n, 0) + \sin \theta(0, \dots, 0, 1)$ with $\sum_i z_i^2 = 1$ and $0 < \theta \leq \pi$, and the same is true for the $2n$ -cell of $\mathbb{CP}^\infty$. From cellular cohomology it then follows that the map $L \rightarrow \mathbb{CP}^\infty$ induces isomorphisms on even-dimensional cohomology with $\mathbb{Z}_m$ coefficients. Since $H^*(\mathbb{CP}^\infty; \mathbb{Z}_m)$ is a polynomial ring, we deduce that if $\gamma \in H^2(L; \mathbb{Z}_m)$ is a generator, then γ^k generates $H^{2k}(L; \mathbb{Z}_m)$ for all k .

By Example 3E.1 there is a generator $x \in H^1(L; \mathbb{Z}_m)$ with $\beta(x) = \gamma$. The product formula $(*)$ gives $\beta(x\gamma^k) = \beta(x)\gamma^k - x\beta(\gamma^k) = \gamma^{k+1}$. Thus β takes $x\gamma^k$ to a generator, hence $x\gamma^k$ must be a generator of $H^{2k+1}(L; \mathbb{Z}_m)$. This completely determines the cup product structure in $H^*(L; \mathbb{Z}_m)$ if m is odd since the commutativity property of cup product implies that $x^2 = 0$ in this case. The result is that $H^*(L; \mathbb{Z}_m) \approx \Lambda_{\mathbb{Z}_m}[x] \otimes \mathbb{Z}_m[\gamma]$ for odd m . When m is even this statement needs to be modified slightly by inserting the relation that x^2 is the unique element of order

2 in $H^2(L; \mathbb{Z}_m) \approx \mathbb{Z}_m$, as we showed in Example 3.9 by an explicit calculation in the 2-skeleton of L .

The cup product structure in finite-dimensional lens spaces follows from this since a finite-dimensional lens space embeds as a skeleton in an infinite-dimensional lens space, and the homotopy type of an infinite-dimensional lens space is determined by its fundamental group since it is a $K(\pi, 1)$. It follows that the cup product structure on a lens space $S^{2n+1}/\mathbb{Z}_m$ with $\mathbb{Z}_m$ coefficients is obtained from the preceding calculation by truncating via the relation $y^{n+1} = 0$.

The relation $\beta = \rho\tilde{\beta}$ implies that $\beta^2 = \rho\tilde{\beta}\rho\tilde{\beta} = 0$ since $\tilde{\beta}\rho = 0$ in the long exact sequence containing $\tilde{\beta}$. Because $\beta^2 = 0$, the groups $H^n(X; \mathbb{Z}_m)$ form a chain complex with the Bockstein homomorphisms β as the ‘boundary’ maps. We can then form the associated *Bockstein cohomology groups* $\text{Ker } \beta / \text{Im } \beta$, which we denote $BH^n(X; \mathbb{Z}_m)$ in dimension n . The most interesting case is when m is a prime p , so we shall assume this from now on.

Proposition 3E.3. *If $H_n(X; \mathbb{Z})$ is finitely generated for all n , then the Bockstein cohomology groups $BH^n(X; \mathbb{Z}_p)$ are determined by the following rules:*

- (a) *Each $\mathbb{Z}$ summand of $H^n(X; \mathbb{Z})$ contributes a $\mathbb{Z}_p$ summand to $BH^n(X; \mathbb{Z}_p)$.*
- (b) *Each $\mathbb{Z}_{p^k}$ summand of $H^n(X; \mathbb{Z})$ with $k > 1$ contributes $\mathbb{Z}_p$ summands to both $BH^{n-1}(X; \mathbb{Z}_p)$ and $BH^n(X; \mathbb{Z}_p)$.*
- (c) *A $\mathbb{Z}_p$ summand of $H^n(X; \mathbb{Z})$ gives $\mathbb{Z}_p$ summands of $BH^{n-1}(X; \mathbb{Z}_p)$ and $BH^n(X; \mathbb{Z}_p)$ with β an isomorphism between these two summands, hence there is no contribution to $BH^*(X; \mathbb{Z}_p)$.*

Proof: We will use the algebraic notion of *minimal chain complexes*. Suppose that C is a chain complex of free abelian groups for which the homology groups $H_n(C)$ are finitely generated for each n . Choose a splitting of each $H_n(C)$ as a direct sum of cyclic groups. There are countably many of these cyclic groups, so we can list them as $G_1, G_2, \dots$. For each G_i choose a generator g_i and define a corresponding chain complex $M(g_i)$ by the following prescription. If g_i has infinite order in $G_i \subset H_{n_i}(C)$, let $M(g_i)$ consist of just a $\mathbb{Z}$ in dimension n_i , with generator z_i . On the other hand, if g_i has finite order k in $H_{n_i}(C)$, let $M(g_i)$ consist of $\mathbb{Z}$ ’s in dimensions n_i and $n_i + 1$, generated by x_i and y_i respectively, with $\partial y_i = kx_i$. Let M be the direct sum of the chain complexes $M(g_i)$. Define a chain map $\sigma: M \rightarrow C$ by sending z_i and x_i to cycles ζ_i and ξ_i representing the corresponding homology classes g_i , and y_i to a chain η_i with $\partial \eta_i = k\xi_i$. The chain map σ induces an isomorphism on homology, hence also on cohomology with any coefficients, by Corollary 3.4. The dual cochain complex M^* obtained by applying $\text{Hom}(-, \mathbb{Z})$ splits as the direct sum of the dual complexes $M^*(g_i)$. So in cohomology with $\mathbb{Z}$ coefficients the dual basis element z_i^* generates a $\mathbb{Z}$ summand in dimension n_i , while y_i^* generates a $\mathbb{Z}_k$ summand in dimension $n_i + 1$ since $\delta x_i^* = ky_i^*$. With $\mathbb{Z}_p$ coefficients, p prime, z_i^* gives a $\mathbb{Z}_p$ summand of

$H^{n_i}(M; \mathbb{Z}_p)$, while x_i^* and y_i^* give $\mathbb{Z}_p$ summands of $H^{n_i}(M; \mathbb{Z}_p)$ and $H^{n_i+1}(M; \mathbb{Z}_p)$ if p divides k and otherwise they give nothing.

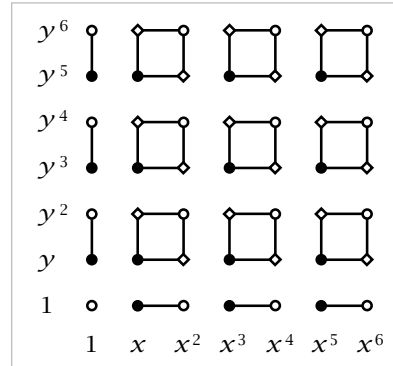
The map σ induces an isomorphism between the associated Bockstein long exact sequences of cohomology groups, with commuting squares, so we can use M^* to compute β and $\tilde{\beta}$, and we can do the calculation separately on each summand $M^*(g_i)$. Obviously β and $\tilde{\beta}$ are zero on y_i^* and z_i^* . When p divides k we have the class $x_i^* \in H^{n_i}(M; \mathbb{Z}_p)$, and from the definition of Bockstein homomorphisms it follows that $\tilde{\beta}(x_i^*) = (k/p)y_i^* \in H^{n_i+1}(M; \mathbb{Z})$ and $\beta(x_i^*) = (k/p)y_i^* \in H^{n_i+1}(M; \mathbb{Z}_p)$. The latter element is nonzero iff k is not divisible by p^2 . $\square$

Corollary 3E.4. *In the situation of the preceding proposition, $H^*(X; \mathbb{Z})$ contains no elements of order p^2 iff the dimension of $BH^n(X; \mathbb{Z}_p)$ as a vector space over $\mathbb{Z}_p$ equals the rank of $H^n(X; \mathbb{Z})$ for all n . In this case $\rho: H^*(X; \mathbb{Z}) \rightarrow H^*(X; \mathbb{Z}_p)$ is injective on the p -torsion, and the image of this p -torsion under ρ is equal to $\text{Im } \beta$.*

Proof: The first statement is evident from the proposition. The injectivity of ρ on p -torsion is in fact equivalent to there being no elements of order p^2 . The equality $\text{Im } \rho = \text{Im } \beta$ follows from the fact that $\text{Im } \beta = \rho(\text{Im } \tilde{\beta}) = \rho(\text{Ker } m)$ in the commutative diagram near the beginning of this section, and the fact that for $m = p$ the kernel of m is exactly the p -torsion when there are no elements of order p^2 . $\square$

Example 3E.5. Let us use Bocksteins to compute $H^*(\mathbb{RP}^\infty \times \mathbb{RP}^\infty; \mathbb{Z})$. This could instead be done by first computing the homology via the general Künneth formula, then applying the universal coefficient theorem, but with Bocksteins we will only need the simpler Künneth formula for field coefficients in Theorem 3.16. The cup product structure in $H^*(\mathbb{RP}^\infty \times \mathbb{RP}^\infty; \mathbb{Z})$ will also be easy to determine via Bocksteins.

For p an odd prime we have $\tilde{H}^*(\mathbb{RP}^\infty; \mathbb{Z}_p) = 0$, hence $\tilde{H}^*(\mathbb{RP}^\infty \times \mathbb{RP}^\infty; \mathbb{Z}_p) = 0$ by Theorem 3.16. The universal coefficient theorem then implies that $\tilde{H}^*(\mathbb{RP}^\infty \times \mathbb{RP}^\infty; \mathbb{Z})$ consists entirely of elements of order a power of 2. From Example 3E.1 we know that Bockstein homomorphisms in $H^*(\mathbb{RP}^\infty; \mathbb{Z}_2) \approx \mathbb{Z}_2[x]$ are given by $\beta(x^{2k-1}) = x^{2k}$ and $\beta(x^{2k}) = 0$. In $H^*(\mathbb{RP}^\infty \times \mathbb{RP}^\infty; \mathbb{Z}_2) \approx \mathbb{Z}_2[x, y]$ we can then compute β via the product formula $\beta(x^m y^n) = (\beta x^m) y^n + x^m (\beta y^n)$. The answer can be represented graphically by the figure to the right. Here the dot, diamond, or circle in the (m, n) position represents the monomial $x^m y^n$ and line segments indicate nontrivial Bocksteins. For example, the lower left square records the formulas $\beta(xy) = x^2 y + x y^2$, $\beta(x^2 y) = x^2 y^2 = \beta(x y^2)$, and $\beta(x^2 y^2) = 0$. Thus in this square we see that $\text{Ker } \beta = \text{Im } \beta$, with generators the ‘diagonal’ sum $x^2 y + x y^2$ and $x^2 y^2$. The



same thing happens in all the other squares, so it is apparent that $\text{Ker } \beta = \text{Im } \beta$ except for the zero-dimensional class '1.' By the preceding corollary this says that all nontrivial elements of $\tilde{H}^*(\mathbb{RP}^\infty \times \mathbb{RP}^\infty; \mathbb{Z})$ have order 2. Furthermore, $\text{Im } \beta$ consists of the subring $\mathbb{Z}_2[x^2, y^2]$, indicated by the circles in the figure, together with the multiples of $x^2y + xy^2$ by elements of $\mathbb{Z}_2[x^2, y^2]$. It follows that there is a ring isomorphism

$$H^*(\mathbb{RP}^\infty \times \mathbb{RP}^\infty; \mathbb{Z}) \approx \mathbb{Z}[\lambda, \mu, \nu] / (2\lambda, 2\mu, 2\nu, \nu^2 + \lambda^2\mu + \lambda\mu^2)$$

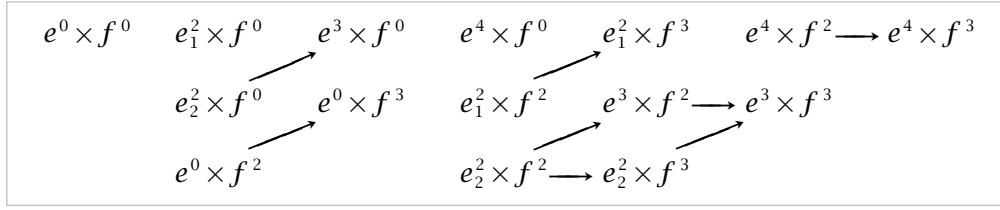
where $\rho(\lambda) = x^2$, $\rho(\mu) = y^2$, $\rho(\nu) = x^2y + xy^2$, and the relation $\nu^2 + \lambda^2\mu + \lambda\mu^2 = 0$ holds since $(x^2y + xy^2)^2 = x^4y^2 + x^2y^4$.

This calculation illustrates the general principle that cup product structures with $\mathbb{Z}$ coefficients tend to be considerably more complicated than with field coefficients. One can see even more striking evidence of this by computing $H^*(\mathbb{RP}^\infty \times \mathbb{RP}^\infty \times \mathbb{RP}^\infty; \mathbb{Z})$ by the same technique.

Example 3E.6. Let us construct finite CW complexes X_1 , X_2 , and Y such that the rings $H^*(X_1; \mathbb{Z})$ and $H^*(X_2; \mathbb{Z})$ are isomorphic but $H^*(X_1 \times Y; \mathbb{Z})$ and $H^*(X_2 \times Y; \mathbb{Z})$ are isomorphic only as groups, not as rings. According to Theorem 3.16 this can happen only if all three of X_1 , X_2 , and Y have torsion in their $\mathbb{Z}$ -cohomology. The space X_1 is obtained from $S^2 \times S^2$ by attaching a 3-cell e^3 to the second S^2 factor by a map of degree 2. Thus X_1 has a CW structure with cells $e^0, e_1^2, e_2^2, e^3, e^4$ with e^3 attached to the 2-sphere $e_0 \cup e_2^2$. The space X_2 is obtained from $S^2 \vee S^2 \vee S^4$ by attaching a 3-cell to the second S^2 summand by a map of degree 2, so it has a CW structure with the same collection of five cells, the only difference being that in X_2 the 4-cell is attached trivially. For the space Y we choose a Moore space $M(\mathbb{Z}_2, 2)$, with cells labeled f^0, f^2, f^3 , the 3-cell being attached by a map of degree 2.

From cellular cohomology we see that both $H^*(X_1; \mathbb{Z})$ and $H^*(X_2; \mathbb{Z})$ consist of $\mathbb{Z}$'s in dimensions 0, 2, and 4, and a $\mathbb{Z}_2$ in dimension 3. In both cases all cup products of positive-dimensional classes are zero since for dimension reasons the only possible nontrivial product is the square of the 2-dimensional class, but this is zero as one sees by restricting to the subcomplex $S^2 \times S^2$ or $S^2 \vee S^2 \vee S^4$. For the space Y we have $H^*(Y; \mathbb{Z})$ consisting of a $\mathbb{Z}$ in dimension 0 and a $\mathbb{Z}_2$ in dimension 3, so the cup product structure here is trivial as well.

With $\mathbb{Z}_2$ coefficients the cellular cochain complexes for X_i , Y , and $X_i \times Y$ are all trivial, so we can identify the cells with a basis for $\mathbb{Z}_2$ cohomology. In X_i and Y the only nontrivial $\mathbb{Z}_2$ Bocksteins are $\beta(e_2^2) = e^3$ and $\beta(f^2) = f^3$. The Bocksteins in $X_i \times Y$ can then be computed using the product formula for β , which applies to cross product as well as cup product since cross product is defined in terms of cup product. The results are shown in the following table, where an arrow denotes a nontrivial Bockstein.



The two arrows from $e_2^2 \times f^2$ mean that $\beta(e_2^2 \times f^2) = e^3 \times f^2 + e_2^2 \times f^3$. It is evident that $BH^*(X_i \times Y; \mathbb{Z}_2)$ consists of $\mathbb{Z}_2$'s in dimensions 0, 2, and 4, so Proposition 3E.3 implies that the nontorsion in $H^*(X_i \times Y; \mathbb{Z})$ consists of $\mathbb{Z}$'s in these dimensions. Furthermore, by Corollary 3E.4 the 2-torsion in $H^*(X_i \times Y; \mathbb{Z})$ corresponds to the image of β and consists of $\mathbb{Z}_2 \times \mathbb{Z}_2$'s in dimensions 3 and 5 together with $\mathbb{Z}_2$'s in dimensions 6 and 7. In particular, there is a $\mathbb{Z}_2$ corresponding to $e^3 \times f^2 + e_2^2 \times f^3$ in dimension 5. There is no p -torsion for odd primes p since $H^*(X_i \times Y; \mathbb{Z}_p) \approx H^*(X_i; \mathbb{Z}_p) \otimes H^*(Y; \mathbb{Z}_p)$ is nonzero only in even dimensions.

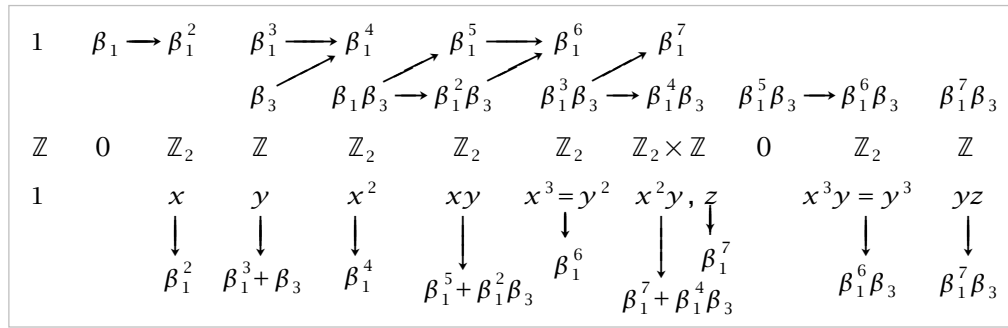
We can see now that with $\mathbb{Z}$ coefficients, the cup product $H^2 \times H^5 \rightarrow H^7$ is nontrivial for $X_1 \times Y$ but trivial for $X_2 \times Y$. For in $H^*(X_i \times Y; \mathbb{Z}_2)$ we have, using the relation $(a \times b) \smile (c \times d) = (a \smile c) \times (b \smile d)$ which follows immediately from the definition of cross product,

- (1) $e_1^2 \times f^0 \smile e_1^2 \times f^3 = (e_1^2 \smile e_1^2) \times (f^0 \smile f^3) = 0$ since $e_1^2 \smile e_1^2 = 0$
- (2) $e_1^2 \times f^0 \smile (e^3 \times f^2 + e_2^2 \times f^3) = (e_1^2 \smile e^3) \times (f^0 \smile f^2) + (e_1^2 \smile e_2^2) \times (f^0 \smile f^3) = (e_1^2 \smile e_2^2) \times f^3$ since $e_1^2 \smile e^3 = 0$

and in $H^7(X_i \times Y; \mathbb{Z}_2) \approx H^7(X_i \times Y; \mathbb{Z})$ we have $(e_1^2 \smile e_2^2) \times f^3 = e^4 \times f^3 \neq 0$ for $i = 1$ but $(e_1^2 \smile e_2^2) \times f^3 = 0 \times f^3 = 0$ for $i = 2$.

Thus the cohomology ring of a product space is not always determined by the cohomology rings of the factors.

Example 3E.7. Bockstein homomorphisms can be used to get a more complete picture of the structure of $H^*(SO(n); \mathbb{Z})$ than we obtained in the preceding section. Continuing the notation employed there, we know from the calculation for $\mathbb{RP}^\infty$ in Example 3E.1 that $\beta(\sum_j \alpha_j^{2i-1}) = \sum_j \alpha_j^{2i}$ and $\beta(\sum_j \alpha_j^{2i}) = 0$, hence $\beta(\beta_{2i-1}) = \beta_{2i}$ and $\beta(\beta_{2i}) = 0$. Taking the case $n = 5$ as an example, we have $H^*(SO(5); \mathbb{Z}_2) \approx \mathbb{Z}_2[\beta_1, \beta_3]/(\beta_1^8, \beta_3^2)$. The upper part of the table at the top of the next page shows the nontrivial Bocksteins. Once again two arrows from an element mean 'sum,' for example $\beta(\beta_1 \beta_3) = \beta(\beta_1) \beta_3 + \beta_1 \beta(\beta_3) = \beta_2 \beta_3 + \beta_1 \beta_4 = \beta_1^2 \beta_3 + \beta_1^5$. This Bockstein data allows us to calculate $H^i(SO(5); \mathbb{Z})$ modulo odd torsion, with the results indicated in the remainder of the table, where the vertical arrows denote the map ρ . As we showed in Proposition 3D.3, there is no odd torsion, so this in fact gives the full calculation of $H^i(SO(5); \mathbb{Z})$.

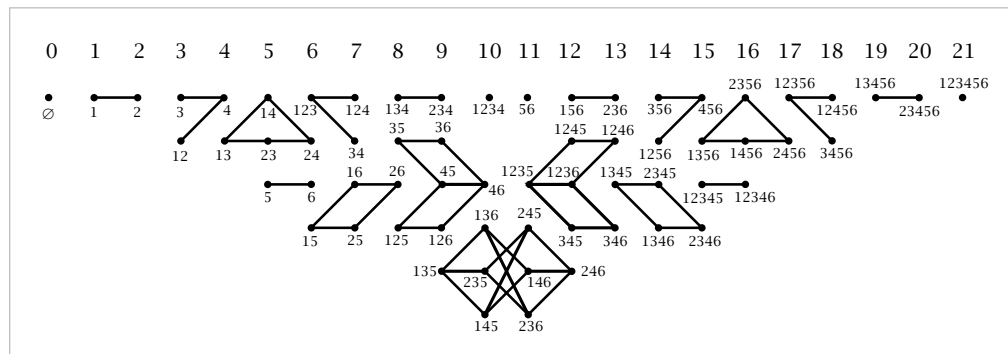


It is interesting that the generator $y \in H^3(SO(5); \mathbb{Z}) \approx \mathbb{Z}$ has y^2 nontrivial, since this implies that the ring structures of $H^*(SO(5); \mathbb{Z})$ and $H^*(\mathbb{RP}^7 \times S^3; \mathbb{Z})$ are not isomorphic, even though the cohomology groups and the $\mathbb{Z}_2$ cohomology rings of these two spaces are the same. An exercise at the end of the section is to show that in fact $SO(5)$ is not homotopy equivalent to the product of any two CW complexes with nontrivial cohomology.

A natural way to describe $H^*(SO(5); \mathbb{Z})$ would be as a quotient of a free graded commutative associative algebra $F[x, y, z]$ over $\mathbb{Z}$ with $|x| = 2$, $|y| = 3$, and $|z| = 7$. Elements of $F[x, y, z]$ are representable as polynomials $p(x, y, z)$, subject only to the relations imposed by commutativity. In particular, since y and z are odd-dimensional we have $yz = -zy$, and y^2 and z^2 are nonzero elements of order 2 in $F[x, y, z]$. Any monomial containing y^2 or z^2 as a factor also has order 2. In these terms, the calculation of $H^*(SO(5); \mathbb{Z})$ can be written

$$H^*(SO(5); \mathbb{Z}) \approx F[x, y, z] / (2x, x^4, y^4, z^2, xz, x^3 - y^2)$$

The next figure shows the nontrivial Bocksteins for $H^*(SO(7); \mathbb{Z}_2)$. Here the numbers across the top indicate dimension, stopping with 21, the dimension of $SO(7)$. The labels on the dots refer to the basis of products of distinct β_i 's. For example, the dot labeled 135 is $\beta_1\beta_3\beta_5$.



The left-right symmetry of the figure displays Poincaré duality quite graphically. Note that the corresponding diagram for $SO(5)$, drawn in a slightly different way from

the preceding figure, occurs in the upper left corner as the subdiagram with labels 1 through 4. This subdiagram has the symmetry of Poincaré duality as well.

From the diagram one can with some effort work out the cup product structure in $H^*(SO(7); \mathbb{Z})$, but the answer is rather complicated, just as the diagram is:

$$F[x, y, z, v, w] / (2x, 2v, x^4, y^4, z^2, v^2, w^2, xz, vz, vw, y^2w, x^3y^2v, \\ y^2z - x^3v, xw - y^2v - x^3v)$$

where x, y, z, v, w have dimensions 2, 3, 7, 7, 11, respectively. It is curious that the relation $x^3 = y^2$ in $H^*(SO(5); \mathbb{Z})$ no longer holds in $H^*(SO(7); \mathbb{Z})$.

Exercises

1. Show that $H^*(K(\mathbb{Z}_m, 1); \mathbb{Z}_k)$ is isomorphic as a ring to $H^*(K(\mathbb{Z}_m, 1); \mathbb{Z}_m) \otimes \mathbb{Z}_k$ if k divides m . In particular, if m/k is even, this is $\Lambda_{\mathbb{Z}_k}[x] \otimes \mathbb{Z}_k[y]$.
2. In this problem we will derive one half of the classification of lens spaces up to homotopy equivalence, by showing that if $L_m(\ell_1, \dots, \ell_n) \simeq L_m(\ell'_1, \dots, \ell'_n)$ then $\ell_1 \cdots \ell_n \equiv \pm \ell'_1 \cdots \ell'_n k^n \pmod{m}$ for some integer k . The converse is Exercise 29 for §4.2.
 - (a) Let $L = L_m(\ell_1, \dots, \ell_n)$ and let $\mathbb{Z}_m^*$ be the multiplicative group of invertible elements of $\mathbb{Z}_m$. Define $t \in \mathbb{Z}_m^*$ by the equation $xy^{n-1} = tz$ where x is a generator of $H^1(L; \mathbb{Z}_m)$, $y = \beta(x)$, and $z \in H^{2n-1}(L; \mathbb{Z}_m)$ is the image of a generator of $H^{2n-1}(L; \mathbb{Z})$. Show that the image $\tau(L)$ of t in the quotient group $\mathbb{Z}_m^* / \pm(\mathbb{Z}_m^*)^n$ depends only on the homotopy type of L .
 - (b) Given nonzero integers $k_1, \dots, k_n$, define a map $\tilde{f}: S^{2n-1} \rightarrow S^{2n-1}$ sending the unit vector $(r_1 e^{i\theta_1}, \dots, r_n e^{i\theta_n})$ in $\mathbb{C}^n$ to $(r_1 e^{ik_1\theta_1}, \dots, r_n e^{ik_n\theta_n})$. Show:
 - (i) $\tilde{f}$ has degree $k_1 \cdots k_n$.
 - (ii) $\tilde{f}$ induces a quotient map $f: L \rightarrow L'$ for $L' = L_m(\ell'_1, \dots, \ell'_n)$ provided that $k_j \ell_j \equiv \ell'_j \pmod{m}$ for each j .
 - (iii) f induces an isomorphism on π_1 , hence on $H^1(-; \mathbb{Z}_m)$.
 - (iv) f has degree $k_1 \cdots k_n$, i.e., f_* is multiplication by $k_1 \cdots k_n$ on $H_{2n-1}(-; \mathbb{Z})$.
 - (c) Using the f in (b), show that $\tau(L) = k_1 \cdots k_n \tau(L')$.
 - (d) Deduce that if $L_m(\ell_1, \dots, \ell_n) \simeq L_m(\ell'_1, \dots, \ell'_n)$, then $\ell_1 \cdots \ell_n \equiv \pm \ell'_1 \cdots \ell'_n k^n \pmod{m}$ for some integer k .
3. Let X be the smash product of k copies of a Moore space $M(\mathbb{Z}_p, n)$ with p prime. Compute the Bockstein homomorphisms in $H^*(X; \mathbb{Z}_p)$ and use this to describe $H^*(X; \mathbb{Z})$.
4. Using the cup product structure in $H^*(SO(5); \mathbb{Z})$, show that $SO(5)$ is not homotopy equivalent to the product of any two CW complexes with nontrivial cohomology.